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Inventor(s)

Rupnow; Calvin T.

Fire Barrier Sliding Mounting System

Abstract

A fire barrier system including a sliding mounting system and being configured to extend across a gap between floor sections is described. The fire barrier system includes a fire barrier fabricated from fire-resistant protective materials draped between the floor sections and a mounting system coupled to the fire barrier and to one of the floor sections. The mounting system includes a mounting rail extending in a lateral direction along the floor section and having a mounting channel, a hook receiver, and a rounded lip. The mounting system includes a retaining bracket coupled to the fire barrier and including a hook end and a rounded end. The retaining bracket is inserted into the mounting channel perpendicular to the lateral direction and rotated into a coupling position within the mounting rail.

Inventors: Rupnow; Calvin T. (Milwaukee, WI)

Applicant: InPro Corporation (Muskego, WI)

Family ID: 1000007747663

Assignee: InPro Corporation (Muskego, WI)

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates generally to the field of fire barriers. The present invention relates specifically to a fire barrier that is installed into floor-to-floor gaps and a sliding mounting system for attaching the fire barrier to sections of flooring.

SUMMARY OF THE INVENTION

[0002] One embodiment of the invention relates to a fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface. The fire barrier system includes a fire barrier fabricated from fire-resistant protective materials. The fire barrier includes a first side, a second side, and a middle section draped between the first side and the second side. The fire barrier system also includes a first mounting system coupled to the first side of the fire barrier and to the first architectural surface. The first mounting system includes a mounting rail coupled to the first architectural surface and extending in a lateral direction along the first architectural surface. The mounting rail includes a mounting channel that extends in the lateral direction, a hook receiver, and a rounded lip. The first mounting system includes a retaining bracket coupled to the first side of the fire barrier, the retaining bracket including a hook end and a rounded end. The fire barrier system includes a second mounting system coupled to the second side of the fire barrier and to the second architectural surface. The retaining bracket is configured to be inserted into the mounting channel from a longitudinal direction that is perpendicular to the lateral direction such that the hook end of the retaining bracket engages the hook receiver of the mounting rail, and the rounded lip of the mounting rail retains the rounded end of the retaining bracket.

[0003] Another embodiment of the invention relates to a fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface. The fire barrier system includes a fire barrier fabricated from fire-resistant protective materials. The fire barrier includes a first side, a second side, and a middle section draped between the first side and the second side. The fire barrier system includes a first mounting system coupled to the first side of the fire barrier and to the first architectural surface. The first mounting system includes a mounting rail coupled to the first architectural surface and extending in a lateral direction along the first architectural surface, the mounting rail including a mounting channel that extends in the lateral direction and a hook receiver defining an upper cavity. The first mounting system includes a retaining bracket coupled to the first side of the fire barrier, the retaining bracket including a hook end. The fire barrier system includes a second mounting system coupled to the second side of the fire barrier and to the second architectural surface. The mounting rail and retaining bracket are configured such that the hook end of the retaining bracket can be inserted into the upper cavity of the hook receiver and the retaining bracket can then rotate with respect to the mounting rail into the mounting channel to couple the retaining bracket to the mounting rail, the retaining bracket rotating about an axis extending in the lateral direction.

[0004] Another embodiment of the invention relates to a method for installing a fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface. The method includes coupling a mounting rail to the first architectural surface such that the mounting rail extends in a lateral direction along the first architectural surface and includes a mounting channel that extends in the lateral direction and a hook receiver defining an upper cavity. The method includes coupling a mounting system to the second architectural surface and coupling a retaining bracket to the mounting rail by inserting a hook end of the retaining bracket into the mounting channel from a longitudinal direction that is perpendicular to the lateral direction, raising the hook end of the retaining bracket into the upper cavity of the mounting rail, and rotating the retaining bracket with respect to the mounting rail into the mounting channel such that the hook end of the retaining bracket engages the hook receiver of the mounting rail to releasably retain the retaining bracket in the mounting channel of the mounting rail, such that the retaining bracket rotates about an axis extending in the lateral direction. The method includes coupling a first side of

a fire barrier to the mounting system and coupling a second side of the fire barrier to the retaining bracket. The fire barrier is fabricated from fire-resistant protective materials and includes a middle section draped between the first side and the second side.

[0005] Additional features and advantages will be set forth in the detailed description which follows, and, in part, will be readily apparent to those skilled in the art from the description or recognized by practicing the embodiments as described in the written description and claims thereof, as well as the appended drawings. It is to be understood that both the foregoing general description and the following detailed description are exemplary.

[0006] The accompanying drawings are included to provide further understanding and are incorporated in and constitute part of the specification. The drawings illustrate one or more embodiment, and together with the description serve to explain the principles and operation of various embodiments.

[0007] Alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] This application will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements in which:

[0009] FIG. 1 is a perspective view of a fire barrier system, according to an exemplary embodiment.

[0010] FIG. 2 is a front side view of the fire barrier system of FIG. 1, according to an exemplary embodiment.

[0011] FIG. 3 is an exploded view of a mounting system of a fire barrier system, according to an exemplary embodiment.

[0012] FIG. 4A is an exploded view of a mounting rail for a fire barrier system placed in a gap between two floor sections, according to an exemplary embodiment.

[0013] FIG. 4B is a detailed view of the mounting rail shown in FIG. 4A, according to an exemplary embodiment.

[0014] FIG. 5A is a perspective view of a mounting rail for a fire barrier system coupled to a first floor section within a gap between the first floor section and a second floor section, according to an exemplary embodiment.

[0015] FIG. 5B is a detailed view of the mounting rail shown in FIG. 5A, according to an exemplary embodiment.

[0016] FIG. 6A is a perspective view of a fire barrier system placed in a gap between a first floor section and a second floor section in which the fire barrier is detached from a mounting rail coupled to the first floor section, according to an exemplary embodiment.

[0017] FIG. 6B is a detailed view of the fire barrier system shown in FIG. 6A, according to an exemplary embodiment.

[0018] FIG. 7A is a detailed view of a mounting system of a fire barrier system in which a retaining bracket of the mounting system is not attached to a mounting rail of the mounting system, according to an exemplary embodiment.

[0019] FIG. 7B is a detailed view of a mounting system of a fire barrier system in which the mounting system is partially assembled, according to an exemplary embodiment.

[0020] FIG. 7C is a detailed view of a mounting system of a fire barrier system in which the mounting system is partially assembled, according to an exemplary embodiment.

[0021] FIG. 7D is a detailed view of a mounting system of a fire barrier system in which the

mounting system is partially assembled, according to an exemplary embodiment.

[0022] FIG. 7E is a detailed view of an assembled mounting system of a fire barrier system, according to an exemplary embodiment.

[0023] FIG. 8 is a detailed view of the fire barrier system of FIG. 1, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0024] Referring generally to the figures, various embodiments of a fire barrier having a fire barrier mounting system and being configured to extend across a gap between architectural surfaces are shown. Embodiments of the fire barrier mounting system discussed herein include an innovative design to provide for a variety of desired characteristics, including releasable attachment of the fire barrier to architectural surfaces, such as floor sections, that allows for relative motion of components of the mounting system in a lateral direction to accommodate lateral displacement between architectural surfaces, and a means of assembling the mounting system without requiring access to the lateral ends of a mounting rail of the mounting system. Typically, traditional fire barrier mounting systems are assembled by attaching a mounting rail to an architectural surface, such that the mounting rail extends laterally across the architectural surface. A frame that is coupled to a fire barrier or blanket is then slid laterally into an end of the mounting rail. In order to assemble the mounting system, clearance of roughly the length of the frame is required from the end of the mounting rail to align the frame with the mounting rail. Such clearance may not be available if the mounting rail is positioned close to a corner or wall connecting to the architectural surface.

[0025] Applicant has found it beneficial to provide a fire barrier mounting system that can be assembled by inserting a portion of a frame or retaining bracket that is coupled to a fire barrier or blanket into an opening in a mounting rail from a longitudinal direction that is perpendicular to the direction the mounting rail extends across. This allows users to easily and quickly assemble and implement the fire barrier without requiring access to the lateral ends of the mounting rail. This also allows users to easily release the fire barrier from the mounting system, for example, to inspect the mounting system or install replacement components.

[0026] Referring to FIG. 1, a fire barrier assembly 2 extending between a first architectural surface 4 and a second architectural surface 6 is shown, according to an exemplary embodiment. Fire barrier assembly 2 includes a fire barrier 8, a first mounting system 10, and a second mounting system 12. First mounting system 10 is coupled to first architectural surface 4 and coupled to fire barrier 8. Second mounting system 12 is coupled to second architectural surface 6 and coupled to fire barrier 8. Both first mounting system 10 and second mounting system 12 extend in a lateral direction 14 along first architectural surface 4 and second architectural surface 6. First architectural surface 4 and second architectural surface 6 are spaced apart from each other in a longitudinal direction 16 that is perpendicular to lateral direction 14. In some embodiments, the distance between first architectural surface 4 and second architectural surface 6 is greater than 6 inches, greater than 20 inches, greater than 24 inches, greater than 30 inches, greater than 40 inches, and/or greater than 50 inches. In a specific embodiment, the distance between first architectural surface 4 and second architectural surface 6 is about 32 inches. First architectural surface 4 and second architectural surface 6 can be floor sections of adjacent building sections.

[0027] As shown in FIG. 2, fire barrier 8 includes a first side 18, a second side 20, and a middle section 22. First side 18 of fire barrier 8 is coupled to first mounting system 10. Second side 20 of fire barrier 8 is coupled to second mounting system 12. Middle section 22 of fire barrier 8 extends between the first side 18 and the second side 20. Fire barrier 8 is formed from a flexible material such that middle section 22 drapes or hangs between first architectural surface 4 and second architectural surface 6. Fire barrier 8 is formed from fire-resistant protective materials. In some embodiments, the fire-resistant protective materials forming fire barrier 8 include one or more insulation blanket layers, intumescent layers, fire-resistant supporting mesh layers, metallic

backing layers, and/or outer protective cloth layers. Fire barrier **8** can be a fire blanket. In the event of a fire, fire barrier **8** is configured to prevent or limit spreading of gases, flames, heat, or smoke through the space between first architectural surface **4** and second architectural surface **6**.

[0028] First mounting system **10** includes a mounting rail **24** and a retaining bracket **26**. Mounting rail **24** is rigidly coupled to first architectural surface **4** by mounting fasteners **28**. Retaining bracket **26** is coupled within mounting rail **24** to attach fire barrier **8** to first architectural surface **4** such that fire barrier **8** extends across the gap between first architectural surface **4** and second architectural surface **6**. In a specific embodiment, mounting rail **24** and/or retaining bracket **26** are formed from aluminum as integrally formed, continuous, contiguous components, respectively, through an extrusion process such that mounting rail **24** and/or retaining bracket **26** are aluminum extrusions.

[0029] Referring to FIG. 2, first mounting system **10** also includes a joint insulator **30**. Joint insulator **30** is coupled to mounting rail **24** and extends below a bottom portion of mounting rail **24** between first architectural surface **4** and fire barrier **8**. Joint insulator **30** is formed from flexible material such that joint insulator **30** can be folded. In a specific embodiment, joint insulator **30** is folded such that a first end **32** of joint insulator **30** and a second end **34** of joint insulator **30** are in contact with mounting rail **24**. When laid flat, joint insulator **30** has a length between first end **32** and second end **34** of between 4 inches and 24 inches. In a specific embodiment, joint insulator **30** has a length of about 12 inches when laid flat. When folded and coupled to mounting rail **24**, joint insulator has a length extending below mounting rail **24** of between 3 and 12 inches. In a specific embodiment, joint insulator **30** has a length extending below mounting rail **24** of about 6 inches when folded and coupled to mounting rail **24**. Joint insulator **30** provides constant pressure to first architectural surface **4** and fire barrier **8** such that gases, flames, or smoke are prevented or limited from passing between first architectural surface **4** and fire barrier **8** in the event of a fire.

[0030] Joint insulator **30** is formed from fire-resistant protective materials. In some embodiments, the fire-resistant protective materials forming joint insulator **30** include a mineral wool layer, an intumescent layer, and a scrim paper layer. In a specific embodiment, the fire-resistant protective materials forming joint insulator **30** includes a plurality of wool layers, intumescent layers, and scrim paper layers. The mineral wool layers are configured to limit heat transfer to mounting rail **24** and retaining bracket **26** in the event of a fire. The intumescent layers are configured to activate when heated and swell, thereby further reducing heat transfer to mounting rail **24** and retaining bracket **26**. The scrim paper layers are configured to provide additional structural integrity to joint insulator **30** and protect the other layers of joint insulator **30**.

[0031] Second mounting system **12** includes a mounting plate **36** that extends into and is coupled to the second side **20** of fire barrier **8**. Mounting plate **36** is rigidly coupled to second architectural surface **6** by mounting fasteners **28**. In some embodiments, second mounting system **12** is substantially similar to first mounting system **10** such that second mounting system **12** includes similar components to first mounting system **10** as discussed herein and is assembled in the same manner as first mounting system **10**.

[0032] Referring to FIG. 3, first mounting system **10** includes a mounting plate **36**, a plurality of fire barrier fasteners **38**, a plurality of washers **40**, and a plurality of nuts **42**. Mounting plate **36** extends into and is coupled to the first side **18** of fire barrier **8**. As shown in FIG. 3, retaining bracket **26** includes a hook end **44**, a rounded end **46**, and a fastener channel **48**. Hook end **44** extends above fastener channel **48** and has a generally arcuate shape. Rounded end **46** extends below fastener channel **48**, opposite hook end **44**, and has a generally arcuate shape. Fastener channel **48** extends in lateral direction **14** along retaining bracket **26** and is configured to receive the heads of fire barrier fasteners **38** to coupled fire barrier fasteners **38** to retaining bracket **26**. Fire barrier fasteners **38** pass through mounting plate **36**, washers **40**, and nuts **42**. Nuts **42** can rotate along threading of fire barrier fasteners **38** to clamp and secure mounting plate **36** and washers **40** to retaining bracket **26**, thereby coupling fire barrier **8** to retaining bracket **26**.

[0033] Mounting rail **24** includes a mounting channel **50**, a hook receiver **52** that defines an upper

cavity **54**, a rounded lip **56**, and a concave arcuate wall **58**. Mounting channel **50** extends along mounting rail **24** in the lateral direction **14** and is configured to receive and releasably retain retaining bracket **26**. Hook receiver **52** extends downward from the top of mounting rail **24** and inward toward first architectural surface **4** such that hook receiver **52** includes a ledge **53**. Upper cavity **54** is located between ledge **53** and the top of mounting rail **24**. Concave arcuate wall **58** is located at the bottom of mounting channel **50** and forms a portion of the perimeter of mounting channel **50**. Rounded lip **56** is located at an end of concave arcuate wall **58** and protrudes upward with respect to the bottom of mounting rail **24**. Concave arcuate wall **58** and rounded lip **56** are configured to releasably retain the rounded end **46** of retaining bracket **26**.

[0034] Referring to FIG. **3**, mounting rail **24** includes a support protrusion **60** and a coupling plate **62**. Support protrusion **60** extends downward from rounded lip **56** toward joint insulator **30**. Coupling plate **62** extends from the bottom of mounting rail **24** and forms a lower portion of mounting rail **24**. In a specific embodiment, coupling plate **62** is coupled to both first end **32** and second end **34** of joint insulator **30** such that coupling plate **62** is sandwiched between portions of joint insulator **30**.

[0035] As shown in FIG. **4A**, first mounting system **10** includes a plurality of mounting fasteners **28** that are spaced apart in the lateral direction **14**. Referring to FIG. **4B**, mounting fasteners **28** are oriented in the longitudinal direction **16**.

[0036] To assemble fire barrier assembly **2**, mounting rail **24** is first placed between first architectural surface **4** and second architectural surface **6**. As shown in FIG. **5A**, mounting rail **24** is then pressed against the first architectural surface **4** such that a surface of mounting rail **24** is flush with first architectural surface **4**. Referring to FIG. **5B**, mounting fasteners **28** are inserted into mounting rail **24** and first architectural surface **4** to couple mounting rail **24** to first architectural surface **4**. Joint insulator **30** is in contact with first architectural surface **4**.

[0037] Referring to FIG. **6A**, fire barrier **8** is then coupled to second architectural surface **6** by second mounting system **12**. Fire barrier **8** extends in the longitudinal direction **16** from second architectural surface **6** toward first architectural surface **4**. As shown in FIG. **6B**, fire barrier **8** is coupled to mounting plate **36** and retaining bracket **26** by a plurality of fire barrier fasteners **38** that are spaced apart from each other in the lateral direction **14**. Retaining bracket **26** is moved toward mounting rail **24** in the longitudinal direction **16** to releasably attach retaining bracket **26** to mounting rail **24** and assemble first mounting system **10**. In an alternative embodiment, fire barrier **8** is coupled to retaining bracket **26** after retaining bracket **26** is coupled to mounting rail **24**.

[0038] Referring to FIGS. **7A-7E**, steps included in a method for assembling first mounting system **10** are shown, according to an exemplary embodiment. As shown in FIGS. **7A** and **7B**, retaining bracket **26** is moved toward mounting rail **24** in the longitudinal direction **16**. Retaining bracket **26** is rotated about an axis that extends in the lateral direction **14** such that hook end **44** is lowered with respect to rounded end **46** and the edge of hook end **44** is angled upward. Retaining bracket **26** is then inserted into mounting rail **24**. Specifically, hook end **44** of retaining bracket **26** is inserted into an opening of mounting channel **50** from the longitudinal direction **16**.

[0039] As shown in FIGS. **7B** and **7C**, hook end **44** is then raised and inserted into upper cavity **54**. Retaining bracket **26** is rotated about an axis extending in the lateral direction **14** such that rounded end **46** is lowered with respect to hook end **44**. Referring to FIGS. **7C** and **7D**, as retaining bracket **26** is rotated, rounded end **46** moves in the longitudinal direction **16** into mounting channel **50**. As shown in FIG. **7D**, when hook end **44** and rounded end **46** of retaining bracket **26** are both within mounting channel **50**, fire barrier **8** is in direct contact with joint insulator **30**.

[0040] Referring to FIGS. **7D** and **7E**, retaining bracket **26** is then lowered within mounting channel **50**. Hook end **44** is lowered within upper cavity **54** until hook end **44** engages with ledge **53** of hook receiver **52**. Rounded end **46** is lowered into contact with concave arcuate wall **58**. Concave arcuate wall **58** has a radius of curvature that is greater than or equal to the radius of curvature of rounded end **46** such that rounded end **46** fits within the curve of concave arcuate wall

58. Concave arcuate wall **58** and rounded lip **56** cooperate to retain rounded end **46** within the curve of concave arcuate wall **58** when downward forces are applied to retaining bracket **26** or when forces are applied to retaining bracket **26** in the longitudinal direction **16** away from mounting rail **24**.

[0041] As shown in FIG. 7E, first mounting system **10** is assembled when retaining bracket **26** is within mounting channel **50** such that hook end **44** engages ledge **53** and/or when rounded end **46** is retained in the curve of concave arcuate wall **58**. Retaining bracket **26** is releasably retained within mounting rail **24** such that first mounting system **10** can be disassembled by reversing the method discussed above. To disassemble first mounting system **10**, retaining bracket **26** can be raised with respect to mounting rail **24** and rotated about an axis extending in the lateral direction **14** such that rounded end **46** is removed from mounting channel **50**. Retaining bracket **26** can then be moved in the longitudinal direction **16** away from mounting rail **24** to remove retaining bracket **26** from mounting channel **50**.

[0042] Fire barrier assembly **2** is shown in an assembled state in FIG. 8, according to an exemplary embodiment. When first mounting system **10** is assembled, retaining bracket **26** extends within mounting rail **24** in the lateral direction **14**. Retaining bracket **26** and mounting rail **24** are configured to allow retaining bracket **26** to slide in the lateral direction **14** with respect to mounting rail **24**. The sliding relationship between retaining bracket **26** and mounting rail **24** accommodates lateral movement of first architectural surface **4** with respect to second architectural surface **6**, such as in response to expansions or contractions from temperature changes or from seismic activity. First mounting system **10** is configured to maintain contact between fire barrier assembly **2** and both first architectural surface **4** and second architectural surface **6** when first architectural surface **4** and second architectural surface **6** move laterally with respect to each other.

[0043] Fire barrier **8**, first mounting system **10**, and second mounting system **12** cooperate to extend across and cover the gap between first architectural surface **4** and second architectural surface **6**. In various embodiments, fire barrier **8** extends a length in the lateral direction **14** of between about 4 feet and about 10 feet. Fire barrier assembly **2** can include a plurality of fire barriers **8** arranged in series along first architectural surface **4** and second architectural surface **6** in the lateral direction **14**. The plurality of fire barriers **8** can be staggered, such as in the longitudinal direction **16**, to limit heat transfer in the gaps between fire barriers **8**.

[0044] It should be understood that the figures illustrate the exemplary embodiments in detail, and it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0045] Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. The construction and arrangements, shown in the various exemplary embodiments, are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. Some elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process, logical algorithm, or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present invention.

[0046] For purposes of this disclosure, the term “about,” when referring to a length or distance

(e.g., a length of about 10 inches), means within 10 percent above or below the referenced value (e.g., between 9 and 11 inches).

[0047] For purposes of this disclosure, the term “coupled” means the joining of two components directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional member being attached to one another. Such joining may be permanent in nature or alternatively may be removable or releasable in nature.

[0048] While the current application recites particular combinations of features in the claims appended hereto, various embodiments of the invention relate to any combination of any of the features described herein whether or not such combination is currently claimed, and any such combination of features may be claimed in this or future applications. Any of the features, elements, or components of any of the exemplary embodiments discussed above may be used alone or in combination with any of the features, elements, or components of any of the other embodiments discussed above.

[0049] In various exemplary embodiments, the relative dimensions, including angles, lengths and radii, as shown in the Figures are to scale. Actual measurements of the Figures will disclose relative dimensions, angles and proportions of the various exemplary embodiments. Various exemplary embodiments extend to various ranges around the absolute and relative dimensions, angles and proportions that may be determined from the Figures. Various exemplary embodiments include any combination of one or more relative dimensions or angles that may be determined from the Figures. Further, actual dimensions not expressly set out in this description can be determined by using the ratios of dimensions measured in the Figures in combination with the express dimensions set out in this description. As used herein, the article “a” is intended to include one or more component or element and is not intended to be construed as meaning only one.

Claims

1. A fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface, the fire barrier system comprising: a fire barrier fabricated from a fire-resistant protective material, the fire barrier comprising: a first side; a second side; and a middle section draped between the first side and the second side; a first mounting system coupled to the first side of the fire barrier and to the first architectural surface, the first mounting system comprising: a mounting rail coupled to the first architectural surface and extending in a lateral direction along the first architectural surface, the mounting rail including a mounting channel that extends in the lateral direction, a hook receiver, and a rounded lip; and a retaining bracket coupled to the first side of the fire barrier, the retaining bracket including a hook end and a rounded end; a second mounting system coupled to the second side of the fire barrier and to the second architectural surface; wherein the retaining bracket is configured to be inserted into the mounting channel from a longitudinal direction that is perpendicular to the lateral direction such that the hook end of the retaining bracket engages the hook receiver of the mounting rail, and the rounded lip of the mounting rail retains the rounded end of the retaining bracket.
2. The fire barrier system of claim 1, wherein, when the retaining bracket is inserted into the mounting channel, the retaining bracket is configured to slide in the lateral direction with respect to the mounting rail.
3. The fire barrier system of claim 1, wherein a distance between the first mounting system and the second mounting system is at least 6 inches.
4. The fire barrier system of claim 1, the first mounting system further comprising a first insulator extending from a bottom portion of the mounting rail, the first insulator comprising: one or more mineral wool layers; one or more scrim paper layers; and one or more intumescent layers.

5. The fire barrier system of claim 4, wherein the first insulator is folded such that the bottom portion of the mounting rail is sandwiched between a first end of the first insulator and a second end of the first insulator.
6. The fire barrier system of claim 1, wherein the mounting rail and the retaining bracket each comprise aluminum and are formed by an aluminum extrusion process.
7. The fire barrier system of claim 1, the first mounting system further comprising: a first fire barrier mounting plate coupled to the fire barrier; a fastener channel formed by the retaining bracket; a plurality of retaining fasteners coupled to the retaining bracket within the fastener channel and coupled to the first fire barrier mounting plate.
8. The fire barrier system of claim 1, further comprising a plurality of mounting fasteners that couple the first mounting system to the first architectural surface and couple the second mounting system to the second architectural surface.
9. A fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface, the fire barrier system comprising: a fire barrier fabricated from fire-resistant protective materials, the fire barrier comprising: a first side; a second side; and a middle section draped between the first side and the second side; a first mounting system coupled to the first side of the fire barrier and to the first architectural surface, the first mounting system comprising: a mounting rail coupled to the first architectural surface and extending in a lateral direction along the first architectural surface, the mounting rail including a mounting channel that extends in the lateral direction and a hook receiver defining an upper cavity; and a retaining bracket coupled to the first side of the fire barrier, the retaining bracket including a hook end; a second mounting system coupled to the second side of the fire barrier and to the second architectural surface; wherein the mounting rail and retaining bracket are configured such that the hook end of the retaining bracket can be inserted into the upper cavity of the hook receiver and the retaining bracket can then rotate with respect to the mounting rail into the mounting channel to couple the retaining bracket to the mounting rail, the retaining bracket rotating about an axis extending in the lateral direction.
10. The fire barrier system of claim 9, the mounting rail further comprising a concave arcuate wall forming a portion of a perimeter of the mounting channel; the retaining bracket further comprising a rounded end opposite the hook end; and wherein the rounded end of the retaining bracket is in contact with the concave arcuate wall of the mounting rail when the retaining bracket is coupled to the mounting rail.
11. The fire barrier system of claim 9, wherein, when the retaining bracket is coupled to the mounting rail, the retaining bracket is configured to slide in the lateral direction with respect to the mounting rail.
12. The fire barrier system of claim 9, wherein a distance between the first mounting system and the second mounting system is at least 6 inches.
13. The fire barrier system of claim 9, the first mounting system further comprising a first insulator extending from a bottom portion of the mounting rail, the first insulator comprising: one or more mineral wool layers; one or more scrim paper layers; and one or more intumescent layers.
14. The fire barrier system of claim 13, wherein the first insulator is folded such that the bottom portion of the mounting rail is sandwiched between a first end of the first insulator and a second end of the first insulator.
15. The fire barrier system of claim 9, wherein the mounting rail and the retaining bracket each comprise aluminum and are formed by an aluminum extrusion process.
16. The fire barrier system of claim 9, the first mounting system further comprising: a first fire barrier mounting plate coupled to the fire barrier; a fastener channel formed by the retaining bracket; a plurality of retaining fasteners coupled to the retaining bracket within the fastener channel and coupled to the first fire barrier mounting plate.
17. The fire barrier system of claim 9, further comprising a plurality of mounting fasteners that couple the first mounting system to the first architectural surface and couple the second mounting

system to the second architectural surface.

18. A method for installing a fire barrier system configured to cover a gap between a first architectural surface and a second architectural surface, the method comprising: coupling a mounting rail to the first architectural surface, the mounting rail extending in a lateral direction along the first architectural surface and including a mounting channel that extends in the lateral direction and a hook receiver defining an upper cavity; coupling a mounting system to the second architectural surface; coupling a retaining bracket to the mounting rail by inserting a hook end of the retaining bracket into the mounting channel from a longitudinal direction that is perpendicular to the lateral direction, raising the hook end of the retaining bracket into the upper cavity of the mounting rail, and rotating the retaining bracket with respect to the mounting rail into the mounting channel such that the hook end of the retaining bracket engages the hook receiver of the mounting rail to releasably retain the retaining bracket in the mounting channel of the mounting rail, the retaining bracket rotating about an axis extending in the lateral direction; coupling a first side of a fire barrier to the mounting system; coupling a second side of the fire barrier to the retaining bracket; wherein the fire barrier is fabricated from fire-resistant protective materials and includes a middle section draped between the first side and the second side.

19. The method of claim 18, wherein the fire barrier is coupled to the retaining bracket before the retaining bracket is coupled to the mounting rail.

20. The method of claim 18, wherein the retaining bracket is coupled to the mounting rail before the fire barrier is coupled to the retaining bracket.
